

Final Examination - ECE 124

Digital Circuits and Systems - Spring 2022

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Date of exam: Monday, August 8, 2022

Exam start time: 12:30PM. Exam end time: 3:00PM. Exam duration: 2.5 hours

Exam type: Closed book. Materials allowed: None

Instructions:

- You must write your **Student ID** number and **Quest Login** userid in the space provided above.
- Answer all questions in this exam. Clearly state any assumptions you make when answering any questions. Show your work—no magic answers are acceptable.
- Both sides of each page are numbered. Your answer to a question can continue to the next numbered page only. For example, your answer to the question on p. 3 can continue to p. 4 only. Failure to follow this instruction properly can cause the answer to be not graded.

Properties of Boolean algebra (only a few of these are needed for answering questions in the exam):

- | | | |
|---|---|----------------|
| 1. (a) $x \cdot 0 = 0$ | (b) $x + 1 = 1$ | |
| 2. (a) $x \cdot 1 = x$ | (b) $x + 0 = x$ | |
| 3. (a) $x \cdot x = x$ | (b) $x + x = x$ | |
| 4. (a) $x \cdot x' = 0$ | (b) $x + x' = 1$ | |
| 5. (a) $(x')' = x$ | | |
| 6. (a) $x \cdot y = y \cdot x$ | (b) $x + y = y + x$ | (commutative) |
| 7. (a) $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ | (b) $x + (y + z) = (x + y) + z$ | (associative) |
| 8. (a) $x \cdot (y + z) = x \cdot y + x \cdot z$ | (b) $x + y \cdot z = (x + y) \cdot (x + z)$ | (distributive) |
| 9. (a) $(x \cdot y)' = x' + y'$ | (b) $(x + y)' = x' \cdot y'$ | (DeMorgan) |
| 10. (a) $x + x \cdot y = x$ | (b) $x \cdot (x + y) = x$ | (absorption) |
| 11. (a) $x \cdot y + x \cdot y' = x$ | (b) $(x + y) \cdot (x + y') = x$ | (combining) |
| 12. (a) $x + x' \cdot y = x + y$ | (b) $x \cdot (x' + y) = x \cdot y$ | |
| 13. (a) $x \cdot y + y \cdot z + x' \cdot z = x \cdot y + x' \cdot z$ | (b) $(x + y) \cdot (y + z) \cdot (x' + z) = (x + y) \cdot (x' + z)$ | (consensus) |

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1. [12 marks]
- Design a 3-bit counter which counts in the sequence:
001, 011, 010, 110, 111, 101, 100, (repeat) 001, . . .
- Fill in the next state columns in the table below. Hint: Use the next expected sequence as the next state as indicated in the table below.
 - Use D flip-flops to implement this counter. Give the DFF inputs in the table below. Draw the circuit diagram.
 - Use T flip-flops to implement this counter. Give the TFF inputs in the table below. Draw the circuit diagram.

Present State			Next State			DFF			TFF		
c	b	a	C	B	A	D _C	D _B	D _A	T _C	T _B	T _A
0	0	0	d	d	d						
0	0	1	0	1	1						
0	1	1									
0	1	0									
1	1	1									
1	1	1									
1	0	1									
1	0	0									

Note: d = Don't care; a, b and c are present state variables and A, B and C are the corresponding next state variables.

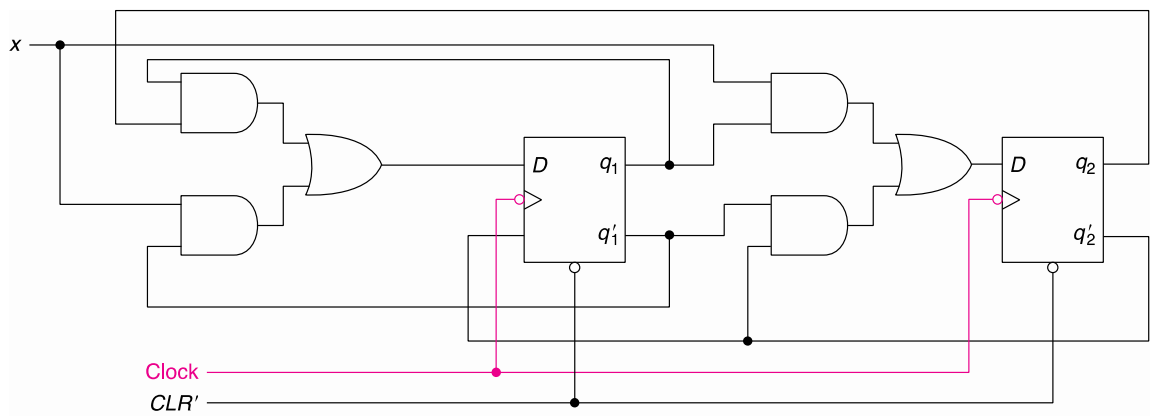
[If needed, use this empty space only to continue with your answer to the question on the previous page.]

2. [8 marks]

Derive a state table for a single-input and single-output Moore-type synchronous FSM that produces an output of 1 if in the input sequence it detects either 110 or 101 patterns. Overlapping sequences should be detected.

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

3. [8 marks]
- Find the state table and the state diagram for the following circuit. Consider q_2 to be the output of the circuit. Ignore the CLR' input.



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4. [8 marks]
- Show the state diagram and the state table of a synchronous Mealy system with an input x and whose output z is 1 if and only if the last four inputs x were 1100 or both of the following conditions are met:
- i) The last output was a 1, and
 - ii) It is continuing on the pattern (1100), i.e., the pattern 1100 is repeating.

The sequence below indicates the relation between the input x and the output z .

x	1	0	1	1	0	0	1	0	1	1	0	0	1	1	0	0	1	1	1	0	0	1	0			
z	?	?	0	0	0	1	1	0	0	0	0	1	1	1	1	1	1	1	0	0	1	1	0	0	0	0

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5. [8 marks]

For a four variable function $f(w_1, w_2, w_3, w_4)$, the initial prime implicant cover table is given below, where prime implicants are represented as patterns of 0, 1 and x, e.g., $p_1 = 00xx = w_1'w_2'$. As it can be seen in the table, there are no *essential* prime implicants for the function. Also, there are no dominant rows or columns in the initial cover table. Apply Petrick's method to find a minimum-cost SOP expression for the function.

Prime implicants	Minterms			
	0	3	10	15
$p_1=00xx$	✓	✓		
$p_2=xx11$		✓		✓
$p_3=x01x$		✓	✓	
$p_4=1x1x$			✓	✓
$p_5=x0x0$	✓		✓	

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

6. [6 marks]

A full adder has three inputs x_i , y_i and c_i and two outputs s_i and c_{i+1} . These symbols have their usual meaning.

- a) For output s_i , give the truth table, a logic expression and a circuit diagram with the *least* number of gates. You are allowed to use NOT, AND, OR and XOR gates with any number of inputs.
- b) For output c_{i+1} , it is given that $c_{i+1} = g_i + p_i c_i$, where $g_i = x_i y_i$, $p_i = x_i \oplus y_i$, and $\oplus$ is XOR. Write an SOP expression for c_3 in terms of c_0 , g_i and p_i , for $0 \leq i \leq 2$.

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

7. [6 marks]

Consider two n -bit 2's complement numbers: $X = x_{n-1}x_{n-2} \dots x_0$ and $Y = y_{n-1}y_{n-2} \dots y_0$. Let the sum of X and Y be $S = s_{n-1}s_{n-2} \dots s_0$. If X and Y are of the same sign and S be of opposite sign, express c_n in terms of c_{n-1} only. (To answer this question, you may use the logic expressions for s_i and c_{i+1} from question 6.)

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

8. [8 marks]
- Apply the partitioning procedure to identify *equivalent* states in the flow table below. (To answer this question, you do not need to explicitly construct a reduced flow table.)

Present state	Next state					Output
	$w_2w_1 =$	00	01	10	11	z
A		B	Ⓐ	-	H	1
B		Ⓑ	F	C	-	0
C		G	-	Ⓒ	D	0
D		-	F	-	Ⓓ	1
E		G	-	Ⓔ	D	1
F		-	Ⓕ	-	K	0
G		Ⓖ	A	J	-	0
H		-	L	E	Ⓗ	1
J		G	-	Ⓙ	-	0
K		-	A	E	Ⓚ	1
L		B	Ⓛ	-	K	1

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

9. [8 marks]

Identify *compatible* states in the following flow table, draw the corresponding merger diagram and give the reduced flow table. (You do not need to apply the merging procedure multiple times.)

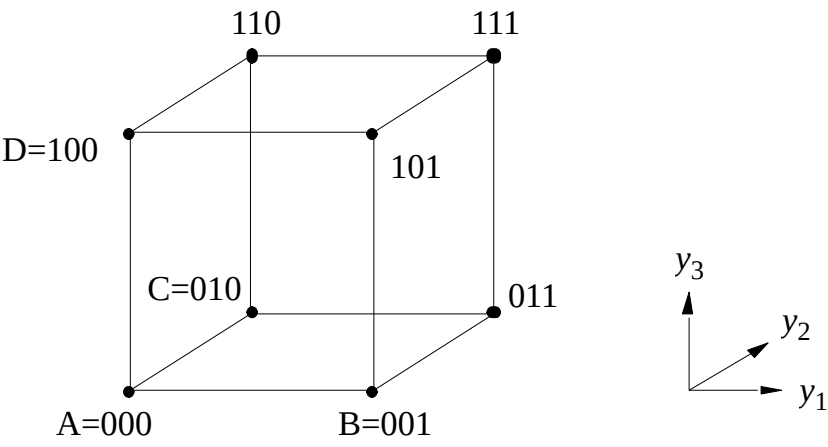
Present state	Next state				Output	
	$w_2w_1 =$	00	01	10	11	z
A		(A)	G	E	-	1
B		F	-	(B)	D	0
C		F	(C)	-	H	0
D		-	C	E	(D)	1
E		A	-	(E)	D	1
F		(F)	C	B	-	0
G		F	(G)	-	D	0
H		-	-	E	(H)	1

[If needed, use this empty space only to continue with your answer to the question on the previous page.]

10. [8 marks]

Below is a flow table for an asynchronous FSM. Find a race-free assignment for the states by using *extra unstable states* and embedding them in a 3-D cube. The cube along with its corner coordinates is shown below. For embedding, assume that the coordinates 000, 001, 010 and 100 have been assigned to states A, B, C and D, respectively (see the diagram below). Use the remaining coordinates as necessary for your extra unstable states such that transition from one stable state to another requires *no more than one unstable state*. Give the modified flow table.

Present state	Next state				Output
	$w_2w_1 =$	00	01	10	z_2z_1
A		B	(A)	D	(A)
B		(B)	(B)	D	A
C		(C)	A	(C)	D
D		C	B	(D)	(D)



[If needed, use this empty space only to continue with your answer to the question on the previous page.]